

Component importance measures by linear programming bounds on system reliability

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ABSTRACT: In order to improve the reliability of civil infrastructure systems against deterioration and natural or man-made hazards, it is important to identify their critical components and cut sets. There exist various measures of component importance, which often relate to the relative contribution of a component or cut set to the failure probability of the system. Recently, the authors developed a linear programming (LP) approach for computing bounds on the failure probability of general systems in terms of the marginal and joint component failure probabilities. This paper shows that bounds on the well-known importance measures can be obtained by solving an LP that is modified or parameterized from the one used for estimating the system failure probability. Also introduced is a new component importance measure that estimates the upgrading of system performance that can be achieved by a fixed cost. The new measure involves the sensitivities of system failure probability with respect to the marginal and joint component failure probabilities, which are byproducts of the LP for the system failure probability. Example applications to an electrical substation system demonstrate the methods proposed for the existing and the new importance measures.

1 INTRODUCTION

The quality of life of people throughout the world depends on safe and reliable operation of civil infrastructure systems, such as ports and harbors, highway transportation networks, communication systems, and water, gas and power supply systems. One important objective in assessing the reliability of such systems is the identification of critical components and cut sets. These are usually defined as components or cut sets which make significant contributions to the system failure probability for a specified system performance criterion and load hazard. When the system reliability is to be upgraded by retrofitting or strengthening of the components, the identified critical components should have a priority over other components, especially when the system upgrade is subject to cost or time constraints.

Various importance measures (IM) have been defined for evaluating and ranking the contributions of components and cut sets to the failure probability of a system. When the component failure events are statistically independent of each other, these IM's can be easily computed in terms of the marginal component failure probabilities (Henley and Kumamoto 1981; Anders 1990; Der Kiureghian et al.

2005). However, when there is dependence between the component states, it is a daunting task to compute these measures, let alone the system failure probability. This task is even more difficult when complete probability information on joint component failure events is unavailable.

Recently, the authors proposed a method for computing bounds on the failure probability of general systems (Song and Der Kiureghian 2003a). This method estimates the bounds in terms of any partial list of marginal and joint component failure probabilities by use of linear programming (LP). In this paper, this approach is used to evaluate a set of well-known IM's by solving an LP, which is a modified or parameterized version of the LP used for computing the system failure probability. Inheriting all the advantages of the LP bounding approach, this evaluation is not hindered by statistical dependence between the component failure states and can accommodate any type of probability information, including inequality-type constraints on individual or joint component failure probabilities.

The importance of components in a system has also been measured in terms of the sensitivity of the system failure probability with respect to the marginal failure probabilities of the individual components (Henley and Kumamoto 1981; Borgonovo and Apostolakis 2001). This 'marginal failure sensitiv-

ity' is useful in estimating the increase in the system reliability that can be achieved by changes in the parameters of a component. When there is dependence between the component states, however, the system failure probability cannot be described by the marginal failure probabilities alone. Hence, the 'marginal failure sensitivity' is useless in estimating the upgrade in the system reliability when the component states are statistically dependent.

We show in this paper that the LP used for estimating bounds on the system failure probability also provides the sensitivities of the system failure probability with respect to the marginal and joint component failure probabilities as byproducts. Using the chain rule together with the sensitivities of the marginal and joint failure probabilities with respect to the component parameters, one can estimate the change in the system reliability for any given set of small changes in the component parameters.

However, one cannot directly compare these 'parameter sensitivities' in order to determine their relative importances. This is because different parameters may be of different dimensions, or that their variations may be associated with different costs. Even though the system failure probability is highly sensitive to a component, high cost of upgrading the component may compromise its importance, especially when the upgrading budget is limited. To address this problem, a new importance measure is defined that relates to the increase in the system reliability that can be achieved by a change in the component parameter at a fixed cost.

As an area of application, in this paper we consider the component importance measures of an electrical substation system. Bounds on several well-known IM's are estimated by use of the proposed parameterized LP approach. Furthermore, the new parameter importance measures are computed for the mean capacities of the components of the system, thus identifying the relative importance of the components for a fixed cost upgrade of the system.

2 SYSTEM RELIABILITY BOUNDS BY LINEAR PROGRAMMING

Song and Der Kiureghian (2003a) have shown that LP can be used to obtain the narrowest possible bounds on the failure probability of a general system for any given partial information on marginal and joint component failure probabilities. The lower (upper) bound on the system failure probability is obtained by solving the LP problem

$$\text{minimize (maximize) } \mathbf{c}^T \mathbf{p} \quad (1a)$$

$$\text{subject to } \mathbf{A}_1 \mathbf{p} = \mathbf{b}_1 \quad (1b)$$

$$\mathbf{A}_2 \mathbf{p} \geq \mathbf{b}_2 \quad (1c)$$

$$\mathbf{A}_3 \mathbf{p} \leq \mathbf{b}_3 \quad (1d)$$

where $\mathbf{p} = \{p_1, p_2, \dots, p_{2^n}\}$ is a vector containing the probabilities of all mutually exclusive and collectively exhaustive (MECE) events in the sample space of the n component states, $\mathbf{c}$ is a vector of binary constants making $\mathbf{c}^T \mathbf{p}$ the probability of the system failure event, and (1b), (1c) and (1d) are equality and inequality constraints expressing known values or bounds on marginal or joint component probabilities, or constraints imposed by the axioms of probability. Note that in (1c) and (1d), the inequalities between the vectors must be interpreted component-wise.

This method has been used to assess the reliability of structural systems and electrical substations (Song and Der Kiureghian 2003a; 2003b), which are usually too complex to be analyzed analytically and often have incomplete probabilistic information. Applications of this approach to complex systems and to reliability updating are described in Der Kiureghian and Song (2005).

3 IMPORTANCE MEASURES RELATIVE TO PARTICIPATION OF COMPONENTS IN SYSTEM FAILURE

3.1 Existing importance measures

This section reviews several well-known IM's that measure the participation of components and cut sets in the system failure event. The Fussell-Vesely (FV) IM measures the fraction of the system failure probability, which is contributed by cut sets containing the component of interest (Fussell 1973). For component i , the FV IM is defined as

$$FV_i = \frac{P(\bigcup_{k: C_k \supseteq E_i} C_k)}{P(E_{\text{sys}})} \quad (2)$$

where C_k denotes the k -th cut set of the system, E_i is the i -th component failure event, and E_{sys} denotes the system failure event. This measure quantifies the contribution of each component to the system failure probability.

The failure of important components tends to increase the failure probability of the system. The Risk Achievement Worth (RAW) IM quantifies the importance of a component by the increase in the system failure probability when the component is removed from the system (Borgonovo and Apostolakis 2001). The RAW of the i -th component is defined as

$$RAW_i = \frac{P(E_{sys}^{(i)})}{P(E_{sys})} \quad (3)$$

where $P(E_{sys}^{(i)})$ denotes the failure probability of the system with component i removed.

The Risk Reduction Worth (RRW) IM measures the decrease in the system failure probability when the component of interest is replaced by a perfectly reliable component (Vinod et al. 2003). Thus, the RRW of the i -th component is

$$RRW_i = \frac{P(E_{sys})}{P(E_{sys}^{(i)})} \quad (4)$$

where $P(E_{sys}^{(i)})$ denotes the failure probability of the system when the i -th component is replaced with a perfectly reliable component. This probability is placed in the denominator of (4) so that a higher value of RRW indicates higher importance of the corresponding component.

The Boundary Probability (BP) of a component measures the change in the probability of failure of the system that is solely due to a change in the state of the component (Anders 1990). It is obtained as

$$BP_i = P(E_{sys}^{(i)}) - P(E_{sys}^{(\bar{i})}) \quad (5)$$

with the two probability terms as defined above.

The critical cut sets can be identified by measuring the contributions of the individual cut sets to the system failure probability. For this purpose, the Fussell-Vesely Cut set (FVC) IM for the k -th cut set is defined as (Fussell 1973)

$$FVC_k = \frac{P(C_k)}{P(E_{sys})} \quad (6)$$

The above IM's require computation of the probabilities of the system failure event and of the various auxiliary system events shown in (2)-(6). In earlier studies, these measures were investigated primarily for systems with independent components. As mentioned earlier, for systems with dependent component states, this is a difficult task not only because of the large computational effort required, but also because of lack of complete probability information. In our earlier study (Song and Der Kiureghian 2004), we used the LP bounding method to approximately compute these IM's. Using this approach, bounds on the BP IM were obtained by simply defining the vector $\mathbf{c}$ in (1a) so the product $\mathbf{c}^T \mathbf{p}$ represents the probability in (5). For the other IM's,

however, one cannot obtain the exact bounds directly by solving an LP because those IM's are not linear functions of $\mathbf{p}$. Therefore, for these IM's, approximate estimates were obtained by substituting the values of $\mathbf{p}$ for lower or upper bounds of each system probability as obtained by the LP approach. It is not guaranteed, however, that these approximate IM's will accurately represent the relative importances of the components or cut sets.

3.2 Importance measures by LP bounds approach

As shown above, the FV, RAW, RRW and FVC IM's have the fractional form

$$IM = \frac{P(A)}{P(B)} \quad (7)$$

where A and B are events in the sample space of the component states. For example, for the FVC IM, A is the k -th cut set C_k and B is the system failure event E_{sys} . We wish to develop a method to compute bounds on the above measure by use of the LP approach. Since events A and B usually are defined as logical functions (involving unions and/or intersections) of the component states of the system, no new variables need to be introduced in the LP formulation.

Recalling that p_i are the probabilities of the disjoint events formed by all possible combinations of the component states, (7) can be written as

$$IM = \frac{\sum_{r \in A} p_r}{\sum_{r \in B} p_r} \quad (8)$$

This is a nonlinear function of $\mathbf{p}$ and LP cannot be directly used to find the desired bounds on this measure. Hence, we consider a reformulation of the problem as follows: Let $P(A) = \mathbf{c}_A^T \mathbf{p}$ and $P(B) = \mathbf{c}_B^T \mathbf{p}$, and consider the parameterized LP problem

$$\text{minimize/maximize } (\mathbf{c}_A^T - \lambda \mathbf{c}_B^T) \mathbf{p} \quad (9a)$$

$$\text{subject to } \mathbf{A}_1 \mathbf{p} = \mathbf{b}_1 \quad (9b)$$

$$\mathbf{A}_2 \mathbf{p} \geq \mathbf{b}_2 \quad (9c)$$

$$\mathbf{A}_3 \mathbf{p} \leq \mathbf{b}_3 \quad (9d)$$

where λ is a parameter to be adjusted. One starts by solving the problem with an initial value λ , usually $\lambda = 1$ for the minimization and $\lambda = 0$ for the maximization. The parameter value is then adjusted according to the rule

$$\lambda = \frac{\mathbf{c}_A^T \hat{\mathbf{p}}}{\mathbf{c}_B^T \hat{\mathbf{p}}} \quad (10)$$

where $\hat{\mathbf{p}}$ is the optimal solution of (9) with the current value of λ . If the objective function of (9) converges to zero at the solution point, the fraction in (10) provides the desired bound. According to Dinkelbach (1967), convergence in this problem is guaranteed and is usually achieved in a few iterations.

It is noteworthy that the proposed method computes the exact bounds on the fraction-type IM's for general systems, regardless of whether statistical dependence between the component events exists or not. Furthermore, these measures can be computed even when the available information on component probabilities is incomplete or is in terms of inequalities.

4 IMPORTANCE MEASURES RELATIVE TO SYSTEM UPGRADE

4.1 Importance measures based on sensitivity

The sensitivities of the failure probability of a system with respect to the marginal failure probabilities of the components are often called Structural Importance Measure (IST; Anders 1990). According to this definition, the IST of the i -th component is

$$IST_i = \frac{\partial P(E_{sys})}{\partial P(E_i)} \quad (11)$$

When the component failure events are statistically independent, the system failure probability is a function of the component marginal failure probabilities and the IST's can be obtained by algebraic manipulations. Furthermore, one can obtain the sensitivity of the system failure probability with respect to a parameter θ by using the chain rule:

$$\frac{\partial P(E_{sys})}{\partial \theta} = \sum_i^n IST_i \frac{\partial P(E_i)}{\partial \theta} \quad (12)$$

However, when there is dependence between the component failure events, the system failure probability cannot be described solely in terms of the marginal failure probabilities. In fact, in such a case, the sensitivities with respect to the marginal failure probabilities may not be well defined. Even if those sensitivities were available, the sensitivities with respect to the joint failure probabilities would also be

required in order to determine the sensitivities with respect to changes in any component parameters.

Most LP algorithms provide the sensitivities of the optimal solution with respect to the constraints of the problem as byproducts (Bertsimas and Tsitsiklis 1997). Thus, when computing the lower or upper bound of the system failure probability by the LP formulation, one can easily obtain the sensitivities of the corresponding system failure probability bound with respect to all available marginal and joint failure probabilities. It is noteworthy that these sensitivities are available even in the cases of statistical dependence between the component failure events, incomplete sets of probabilities or inequality-type constraints.

Using the chain rule of differentiation and employing the sensitivities with respect to all available marginal and joint failure component failures, one can determine the sensitivity of the lower or upper bound of the system failure probability with respect to a parameter θ from

$$\begin{aligned} \frac{\partial P(E_{sys})}{\partial \theta} &\cong \sum_{i \in S_1} \frac{\partial P(E_{sys})}{\partial P(E_i)} \frac{\partial P(E_i)}{\partial \theta} \\ &+ \sum_{(i,j) \in S_2} \frac{\partial P(E_{sys})}{\partial P(E_i E_j)} \frac{\partial P(E_i E_j)}{\partial \theta} + \dots \end{aligned} \quad (13)$$

where S_m denotes the index set of the available m -th order joint failure probabilities. Note that all the terms $\partial P(E_{sys})/\partial P(E_i)$, $\partial P(E_{sys})/\partial P(E_i E_j)$, etc., are obtained as byproducts of the lower or upper bound solution of the LP. Hence, one obtains two approximate solutions of the parameter sensitivity, one from the lower bound solution and one from the upper bound solution.

4.2 Importance measure for fixed-cost upgrading

Here we define a new set of component importance measures that are relevant for a fixed-cost upgrading of the system. The sensitivities in (12) and (13) cannot serve this purpose because different parameters may not be of the same dimension, and also because parameter variations may not be cost neutral. For example, even if the system failure probability is highly sensitive to a component, the component may not rank high in importance if the cost of upgrading it even a small amount is prohibitive. Conversely, a component with lower sensitivity can be important if a large improvement in its capacity can be achieved at low cost, such that the increase in the system reliability is large.

Let $\Delta \theta_i$, $i = 1, \dots, n$, represent an increment in the parameter θ_i of each component that can be achieved at a fixed cost increment. As the measure of importance of a component, we consider the reduction in the system failure probability that can be

achieved by the change $\Delta\theta_i$ in its parameter. Hence, the following importance measure is defined:

$$I_{\theta_i} = -\left[\frac{\partial P(E_{sys})}{\partial \theta_i}\right] \Delta\theta_i \quad (14)$$

This dimensionless measure accounts not only for the influence of the component on the system reliability, but also for the cost of improving the component performance. Note that the measure in (14) is an approximation of the reduction in the system failure probability that results from the upgrading of component i with the fixed cost increment. Hence, the larger this measure is, the more critical the component is. It is noted that when θ_i is used in place of $\Delta\theta_i$, the above measure become identical to the so-called parameter “elasticity” of the system failure probability. Clearly, the measure in (14) is more general, as it accounts for the potential difference in the cost of unit change in different parameters.

5 NUMERICAL EXAMPLES

As a numerical example, consider the seismic reliability of the two-transmission-line substation system described in Song and Der Kiureghian (2003b) and Figure 1. The system consists of 12 interconnected electrical equipment items. It is assumed that the system fails when no connection exists between either of the two input lines and either of the two output lines. Referring to the component numbers indicated in the figure, the system has the 25 minimum cut sets $\{(1,2), (4,5), (4,7), (4,9), (5,6), (6,7), (6,9), (5,8), (7,8), (8,9), (11,12), (1,3,5), (1,3,7), (1,3,9), (2,3,4), (2,3,6), (2,3,8), (4,10,12), (6,10,12), (8,10,12), (5,10,11), (7,10,11), (9,10,11), (1,3,10,12), (2,3,10,11)\}$.

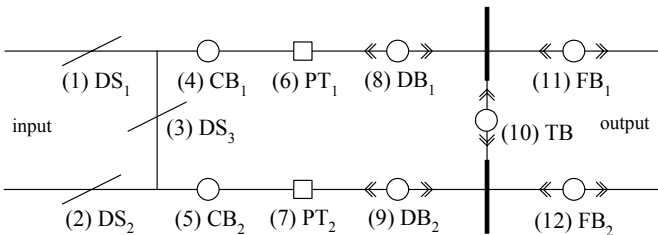


Figure 1. A two-transmission-line substation system

Let R_i denote the capacity of the i -th equipment item with respect to the base acceleration in units of gravity acceleration, g , and assume it has the lognormal distribution. The means and coefficients of variation (c.o.v.) of the equipment capacities are assumed as follows: disconnect switch (DS) $\sim (0.4g, 0.3)$, circuit breaker (CB) $\sim (0.3 g, 0.3)$, power transformer (PT) $\sim (0.5 g, 0.5)$, drawout breaker (DB) $\sim (0.4 g, 0.3)$, feeder breaker (FB) $\sim (1.0 g, 0.3)$ and tie

breaker (TB) $\sim (1.0 g, 0.3)$. The capacities of equipment items within each category are assumed to be equally correlated with a correlation coefficient of 0.3 except 0.5 for PT's. Equipment capacities in different categories are assumed to be statistically independent. The above assumed statistics are rough approximations based on Bayesian analyses of observed data on the performance of electrical substation equipment in past earthquakes (Der Kiureghian 2002).

Let A denote the bed-rock peak ground acceleration (PGA) in the vicinity of the substation during the earthquake and S_i denote a factor representing the local site response for equipment i , such that AS_i is the actual PGA experienced by the i -th equipment item. Assume A is a lognormal random variable with mean $0.15g$ (in units of gravity acceleration, g) and c.o.v. 0.5, and S_i , $i=1,\dots,12$, are independent lognormal random variables, also independent of A , with means 1.0 and c.o.v. 0.2.

The failure events of the individual equipment items are then formulated as

$$E_i = \{\ln R_i - \ln A - \ln S_i \leq 0\} \quad i = 1, \dots, 12 \quad (15)$$

Since the logarithm of a lognormal random variable is normal, $v_i = \ln R_i - \ln A - \ln S_i$ has the normal distribution. Therefore, the marginal failure probability of component i is given by

$$P_i = \Phi\left(-\frac{\mu_i}{\sigma_i}\right) \quad i = 1, \dots, 12 \quad (16)$$

where $\Phi(\cdot)$ denotes the standard normal cumulative distribution function and μ_i and σ_i are the mean and standard deviation of v_i , respectively, which are easily computed in terms of the statistics of A , S_i and R_i given above. Furthermore, for any pair of components i and j , the random variables v_i and v_j are jointly normal and the joint probability of failure of the two components can be computed from (Ditlevsen and Madsen 1996)

$$P_{ij} = \Phi(u_i)\Phi(u_j) + \int_0^{\rho_{ij}} \phi_2(u_i, u_j, \rho) d\rho \quad (17)$$

where $u_i = -\mu_i / \sigma_i$, ρ_{ij} denotes the correlation coefficient between v_i and v_j , and $\phi_2(\cdot, \cdot, \rho)$ represents the bi-variate normal probability density function with zero means, unit standard deviations and correlation coefficient ρ . The joint probability of failure of components i, j and k can be computed by the following double-fold numerical integration by use of conditional normal probabilities and (17).

$$P_{ijk} = \int_{-\infty}^{u_k} \varphi(z) [\Phi(u'_i) \Phi(u'_j) + \int_0^{\rho'} \varphi_2(u'_i, u'_j, \rho) d\rho] dz \quad (18)$$

where $\varphi(\cdot)$ represents the probability density function with zero-mean and unit standard deviation, and u'_i , u'_j and ρ' are defined as

$$u'_i = \frac{u_i - \rho_{ik}z}{\sqrt{1 - \rho_{ik}^2}}, \quad u'_j = \frac{u_j - \rho_{jk}z}{\sqrt{1 - \rho_{jk}^2}} \quad (19a,b)$$

$$\rho' = \frac{\rho_{ij} - \rho_{jk}\rho_{ik}}{\sqrt{1 - \rho_{ik}^2} \sqrt{1 - \rho_{jk}^2}} \quad (19c)$$

The above uni-, bi- and tri-component probabilities are used to compute the bounds on the system failure probability by use of LP in (1). The tri-component bounds are $0.0616 \leq P(E_{sys}) \leq 0.0942$.

5.1 Importance measures relative to participation of components in system failure

The FV IM's of the components are obtained by iteratively solving the parameterized LP in (9). One can use the same constraints used for the system failure probability. In (7), the event A is the union of the cut sets including the component of interest, while the event B is the system failure event. One only needs to define the $\mathbf{c}$ vector for event A . Figure 2 shows the bounds on the FV IM's.

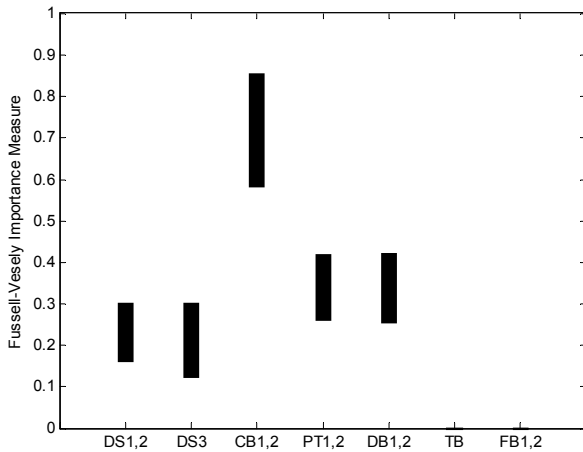


Figure 2. Fussell-Vesely IM

For RAW (resp. RRW) IM, one needs to find the corresponding $\mathbf{c}$ vector for the event $E_{sys}^{(i)}$ (resp. $E_{sys}^{(i)}$). For details on how to find these vectors, see Song (2004). Solving the parameterized LP's with these vectors, RAW and RRW IM's of each component are obtained as shown in Figures 3 and 4.

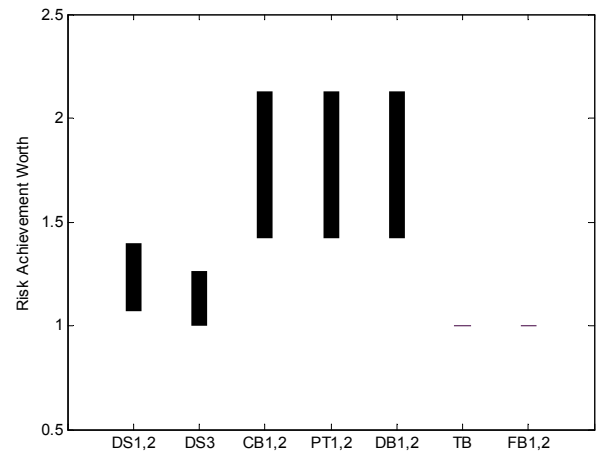


Figure 3. Risk Achievement Worth

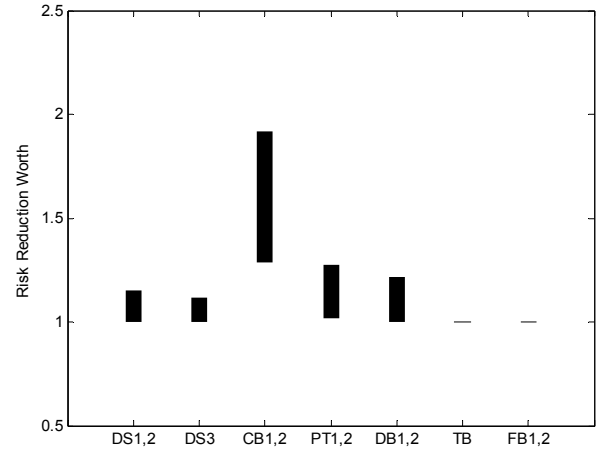


Figure 4. Risk Reduction Worth

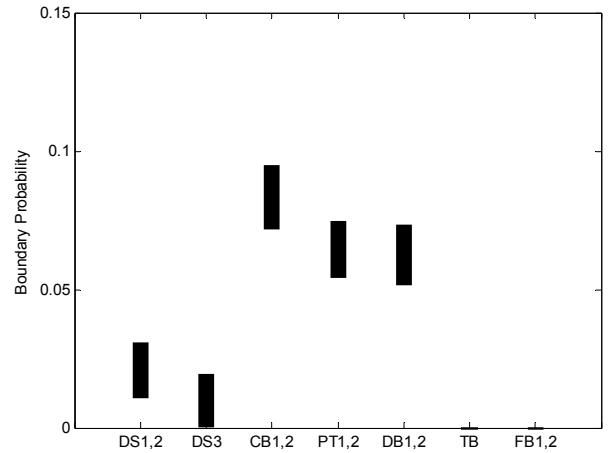


Figure 5. Boundary Probability

For the BP IM, we do not need to solve a parameterized LP. One can compute the BP IM of the i -th component simply by solving the LP in (1) with the vector $\mathbf{c} = \mathbf{c}_{RAW(i)} - \mathbf{c}_{RRW(i)}$, where $\mathbf{c}_{RAW(i)}$ and $\mathbf{c}_{RRW(i)}$ are the objective function vectors for $E_{sys}^{(i)}$ and $E_{sys}^{(i)}$, respectively, and the constraints used for the original system failure probability. Figure 5 shows the BP IM's of the components.

According to the FV, RRW and BP measures, the importance ranking of the components is as follows: $(CB_{1,2}) \rightarrow (PT_{1,2}) \rightarrow (DB_{1,2}) \rightarrow (DS_{1,2}) \rightarrow (DS_3) \rightarrow (FB_{1,2}, TB)$. The ranking according to RAW is $(CB_{1,2}, PT_{1,2}, DB_{1,2}) \rightarrow (DS_{1,2}) \rightarrow (DS_3) \rightarrow (FB_{1,2}, TB)$, which is identical to the order by the other measures, except that CB, PT and DB have the same order of importance. As a result, one can say that, in terms of contributing to the system failure probability, the two circuit breakers ($CB_{1,2}$) are the most critical components in the system followed by the other equipment items as listed above.

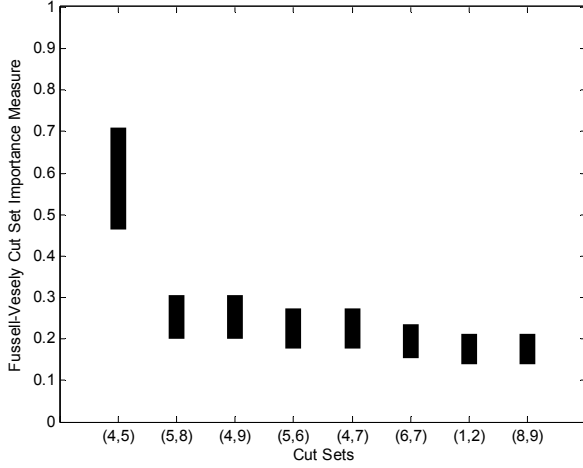


Figure 6. Fussell-Vesely Cut set IM

To identify the critical minimum cut sets, the FVC IM is computed by iteratively solving the parameterized LP in (9) with $\mathbf{c}_A$ and $\mathbf{c}_B$ being the objective function vectors for the cut set of interest and the system failure event, respectively. The cut sets are sorted in the descending order of FVC IM and the first 8 cut sets are shown in Figure 6. The joint failure of the two circuit breakers is the most critical cut set. The joint failures of a drawout breaker (DB) and a circuit breaker on different lines are ranked next. It is noteworthy that the top five most critical cut sets all include at least one circuit breaker. This further reinforces the importance of these components for the system.

5.2 Importance measures relative to system upgrade

We consider the importance measures relative to fixed-cost increments in the logarithmic mean capacities of the equipment items. The sensitivities of the uni-, bi- and tri-component failure probabilities with respect to the parameter $\theta_i = E[\ln R_i]$ are:

$$\frac{\partial P_i}{\partial \theta_i} = \frac{dP_i}{du_i} \frac{\partial u_i}{\partial \theta_i} = \frac{d\Phi(u_i)}{du_i} \left(-\frac{1}{\sigma_i} \right) = -\frac{\phi(u_i)}{\sigma_i} \quad (20)$$

$$\begin{aligned} \frac{\partial P_{ij}}{\partial \theta_i} &= \frac{\partial P_{ij}}{\partial u_i} \frac{\partial u_i}{\partial \theta_i} \\ &= -\frac{1}{\sigma_i} \left[\phi(u_i) \Phi(u_j) + \int_0^{\rho_{ij}} \frac{\partial \phi_2(u_i, u_j, \rho)}{\partial u_i} d\rho \right] \end{aligned} \quad (21)$$

$$\begin{aligned} \frac{\partial P_{ijk}}{\partial \theta_i} &= \frac{\partial P_{ijk}}{\partial u_i} \frac{\partial u_i}{\partial \theta_i} \\ &= -\frac{1}{\sigma_i} \left\{ \phi(u_i) \left[\Phi(u_j'') \Phi(u_k'') + \int_0^{\rho'} \phi_2(u_j'', u_k'', \rho) d\rho \right] \right\} \end{aligned} \quad (22)$$

where

$$u_j'' = \frac{u_j - \rho_{ij} u_i}{\sqrt{1 - \rho_{ij}^2}}, \quad u_k'' = \frac{u_k - \rho_{ik} u_i}{\sqrt{1 - \rho_{ik}^2}}, \quad (23)$$

$$\rho' = \frac{\rho_{jk} - \rho_{ij} \rho_{ik}}{\sqrt{1 - \rho_{ij}^2} \sqrt{1 - \rho_{ik}^2}} \quad (24)$$

The fixed-cost increments in the component capacities are assumed to be $\Delta \theta_i = 1.0, 0.9, 0.3, 1.2, 0.8$, and 1.0 for $DS_{1,2,3}$, $CB_{1,2}$, $PT_{1,2}$, $DB_{1,2}$, TB and $FB_{1,2}$, respectively.

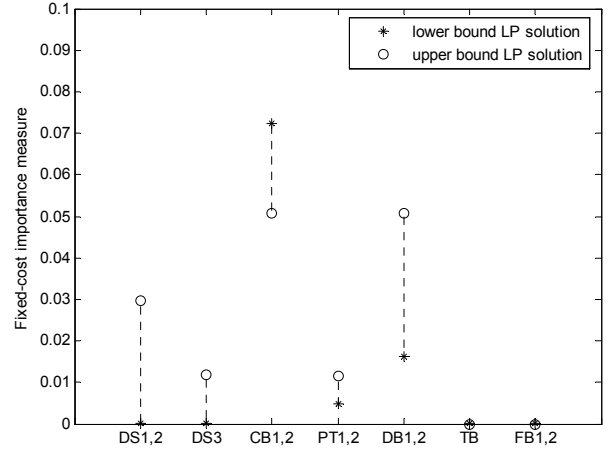


Figure 7. Fixed-cost importance measures

Figure 7 shows the upgrading IM's for fixed-cost increments in the logarithmic means of the component capacities. The two results shown for each component are those obtained based on the lower and upper bounds of the system failure probability. Comparing with the results in Figure 2 for the FV IM's, we see that the order of importance of CB_1 and CB_2 is still the highest, but PT_1 and PT_2 have lost their importance. This is because of the high cost of upgrading these components. Furthermore, DS_1 and DS_2 have gained importance, because of their low upgrading cost.

Two sets of importance measures for the components within a system are defined and algorithms for their computation are developed. One set describes the importance of a component relative to its contribution to the failure probability of the system. Several measures from the literature, which previously have been used primarily for systems with independent components, are employed and a parametric LP formulation is used to compute them for systems with dependent component states.

Next, a new measure of component importance is defined in the context of upgrading the system reliability at a fixed-cost increment. This measure accounts not only for the sensitivity of the system failure probability with respect to the component parameter, but also the cost of upgrading the component reliability. The measure employs sensitivities with respect to the constraints of the LP problem, which are readily available from its solution.

Application to an example electrical substation system demonstrates the use of the component importance measures developed or extended in this paper.

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